

PHYSICS PRACTICAL SHEETS

Date:

Class: BSc 2nd Year

Roll No.: 68

Shift: Day

Object of the Experiment (Block Letter) :

PN CAMPUS

Experiment No.: 11

Group: P₂

Sub.: Physics

Set: 1

TO STUDY THE CHARACTERISTICS OF GEIGER MUELLER (G.M) COUNTER AND ITS RELIABILITY.

APPARATUS REQUIRED:

- (i) A Geiger Mueller counter tube with stand
- (ii) Scalar unit having arrangement for extra high voltage D.C. supply and fitted with a voltmeter
- (iii) A radioactive source (α -source)

THEORY :

Geiger Mueller (GM) Counter: A GM counter tube consists of a glass cylinder 'G' with a metal coating 'M' on the inside on a co-axial wire anode ~~M~~ on the inside on a (A) (usually of tungsten) suspended along its axis. The cylinder is filled with a gas such as argon mixed with halogen vapour at a reduced pressure of about 10 cm of mercury. The tube may have a very thin mica end window W shielded by a protection gauge P, the metal tube is maintained at a negative potential of about 1000 volt with respect to the wire. The wire cylinder potential difference 'V' plotted against the counting rate (CR) owing to a radioactive source of fixed intensity placed near the tube is known as a plateau curve corresponding to the threshold voltage or Geiger Mueller Threshold potential. As the voltage across the tube is

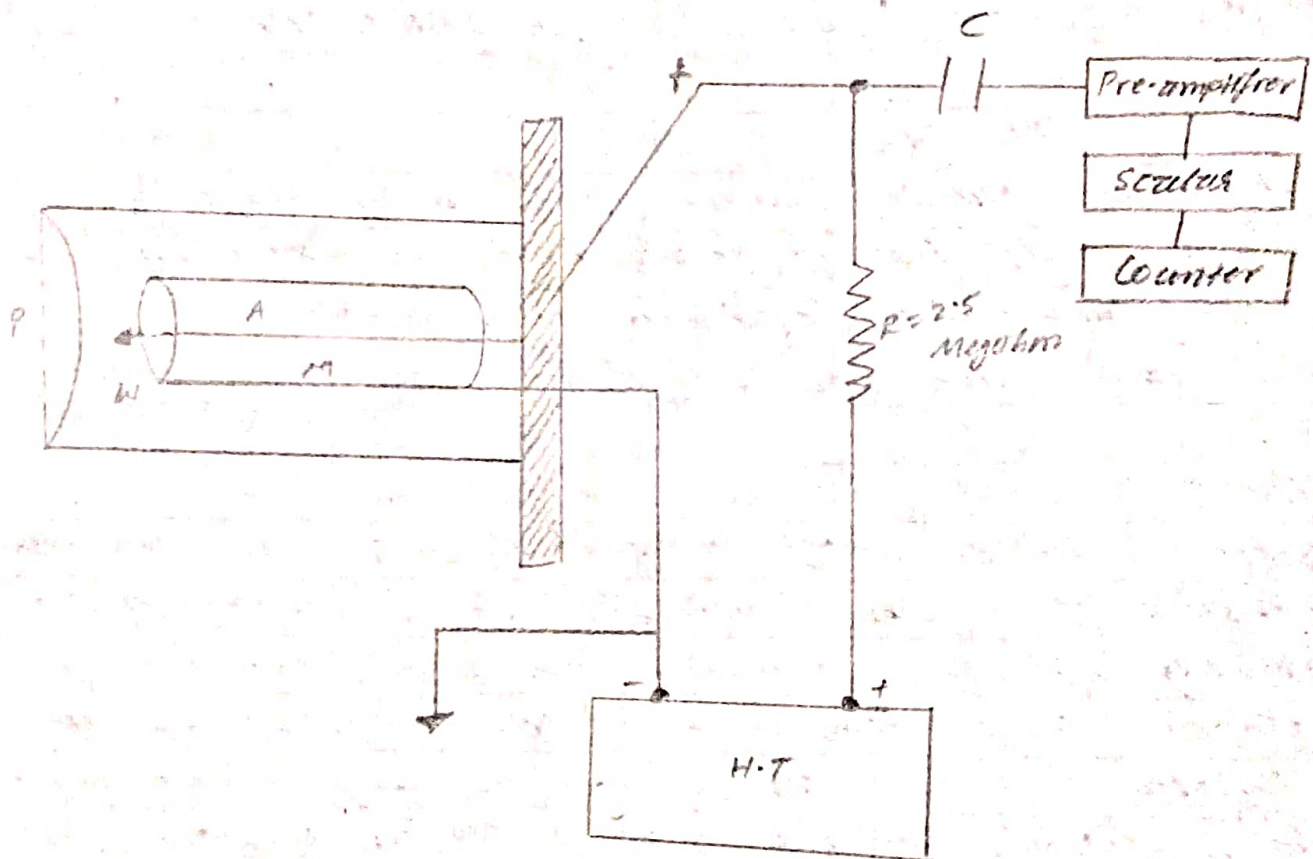


Fig: GM Counter

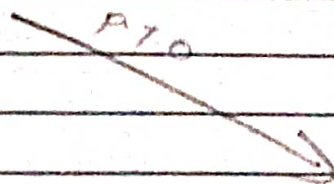
Increased, the counting rate increase rapidly upto a point B (corresponding to voltage V_1) and then levels off to an almost constant value until voltage V_2 corresponding to point 'C' is reached. The portion BC is known as Geiger Mueller region. The voltage from B to C is known as the length of the plateau. The slope of the plateau is the average percentage change in the counting rate per unit increment of applied voltage and is known as relative plateau slope (R.P.S). It is therefore given by,

$$R.P.S = \frac{(N_2 - N_1) / N}{V_2 - V_1} \times 100$$

where N_1 is the counting rate at V_1 and N_2 that at voltage V_2 .

Also,

$$N = \frac{N_1 + N_2}{2}$$



OBSERVATION:

S.N.	Voltage (V)	Background count without source (b)	Count with source (N)	Corrected count rate ($N = N' - b$)
1.	340	0	0	0
2.	360	23	110	87
3.	380	20	101	81
4.	400	16	96	80
5.	420	21	113	92
6.	440	14	125	111
7.	460	19	101	82
8.	480	15	119	104
9.	500	20	104	84
10.	520	26	107	81
11.	540	21 16	101	85
12.	560	25 21	124	103
13.	580	25 25	99	74
14.	600	19 25	93	68
15.	620	33 19	113	94
16.	640	26 33	114	81
17.	660	26	102	76
18.	680	20	112	92
19.	700	27	117	90

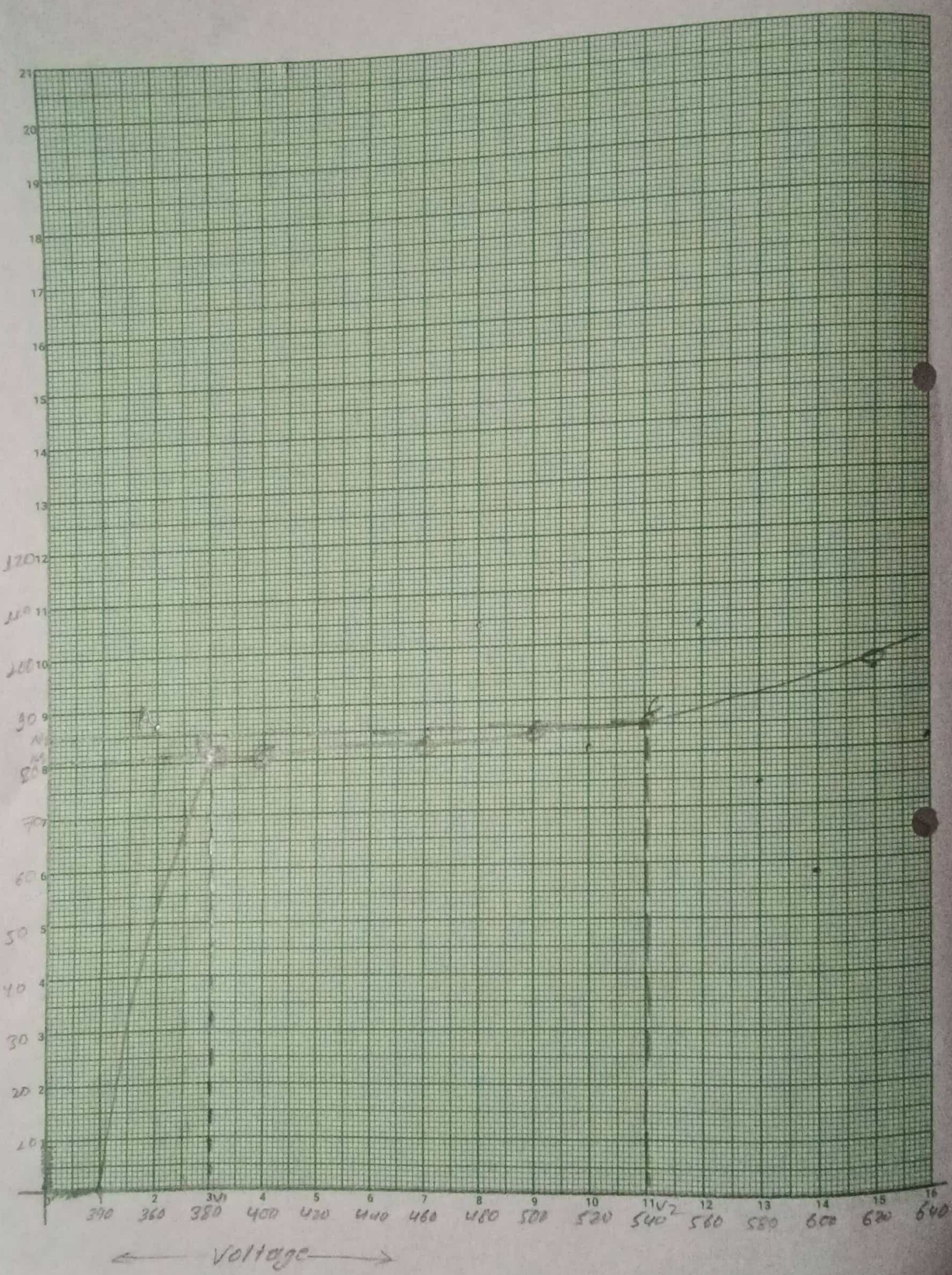
CALCULATION:

From graph,

Threshold voltage = voltage at A = 360 volt

voltage at which plateau begins = voltage at B = V_1 = 380 voltvoltage at which plateau finished = voltage at C = V_2 = 540 voltplateau region = $(V_2 - V_1) = 540 - 380 = 160$ volt

← NO of count of radioactive source →



Operating voltage from the graph,

$$V = V_1 + \frac{1}{2}(V_2 - V_1) = 380 + \frac{160}{2} \\ = 433.33 \text{ volt}$$

Relative Plateau Slope:

Mean corrected counting rate at $V_1 = N_1 = 81$ pulse per min

Mean corrected counting rate at $V_2 = N_2 = 85$ pulse per min

$$\therefore N = (N_1 + N_2) / 2 = (85 + 81) / 2 = 83 \text{ pulse per min}$$

$$\therefore \text{Relative plateau slope (R.P.S)} = \frac{(N_2 - N_1) / N}{(V_2 - V_1)} \times 100\%$$

$$= \frac{(85 - 81) / 83 \times 100\%}{(540 - 380)}$$

$$= 0.03012$$

$$= 3.012 \times 10^{-3} \%$$

RESULT:

Hence, from the above experiment, the relative plateau slope (R.P.S) was found to be $3.012 \times 10^{-3} \%$.

PRECAUTIONS:

- (i) The radioactive material should be handled very carefully by a pair of tongs or forceps.
- (ii) The counting switch should always be kept on so that the counting rate does not exceed 10% above the plateau voltage. Excessive voltage will damage the G.M. counter.
- (iii) The radioactive source should be placed in the source holder just below the Geiger tube before increasing the high voltage slowly.